

Beyond Logistic Regression: Random Forests, XGBoost, and Optuna for NF1 Cell Classification

Mark Talbot

University of Texas at Austin

Austin, Texas, USA

mark.talbot@utexas.edu

Abstract

Neurofibromatosis type 1 (NF1) is caused by loss of function in the NF1 gene and leads to Schwann cell tumors with limited treatment options. Recent work by Tomkinson et al. showed that Cell Painting microscopy features combined with a logistic regression classifier can distinguish NF1 wild type from NF1 null type Schwann cells with high accuracy. In this project, I reproduced their logistic regression model and then extended the machine learning layer with Optuna based hyperparameter tuning, random forests and XGBoost while keeping the imaging and feature extraction pipeline fixed. Logistic regression reproduced the original pipeline performance (.79% test accuracy) and more extensive hyperparameter search did not improve performance. XGBoost achieved a small improvement in test set accuracy (0.80) but showed clear overfitting to the training data and random forests did not outperform logistic regression. When applied to the generalization dataset, all models (including the original paper) performed near chance for MGH cells and only logistic regression showed a modest improvement at iNFixion. These results show that logistic regression remains a strong baseline and that future work should focus on plate artifacts.

1.0 Introduction and Motivation

Neurofibromatosis type 1 (NF1) is a type of genetic disorder caused by loss of function variants in the NF1 gene leading to a deficiency in the production of the tumor suppressor neurofibromin. Neurofibromin normally down regulates the RAS/MAPK signaling so its loss leads to hyperactive signaling and abnormal behavior of Schwann cells. Clinically, this results in multiple tumors along the nerves including cutaneous neurofibromas, plexiform neurofibromas some of which can progress to atypical neurofibromatous neoplasms of uncertain biologic potential (ANNUBP) and ultimately malignant peripheral nerve sheath tumors (MPNST). MPNSTs are highly aggressive with a five year survival rate of 20% to 30% over a five year period. Neurofibromas of all types can be disfiguring and painful which leads to quality of life issues. Neurofibromatosis is relatively common among rare diseases impacting 1 in 3000 individuals [11].

Selumetinib and Mirdametinib are both MEK inhibitors and are the only approved treatments for neurofibromatosis by acting on the RAS/MAPK pathway. There is currently no treatment that addresses the deficiency in neurofibromin. The first step in addressing this would be to develop screening of NF1 deficient cells which would allow the ability to identify therapeutic agents to reduce the neurofibromin deficiency by looking at their phenotype [12].

Image based profiling combined with machine learning is one such route. This would provide the ability to detect NF1 morphologically. The goal is to see how reliably I can classify NF1 +/- vs NF1

-/- from cell painting features and to see how different ML choices impact that reliability.

Tomkinson et al[1] built a logistic regression classifier on cell painting NF1 Schwann cells and developed high accuracy for the trained baseline. Logistic regression is a standard baseline in biological machine learning. The specific question in this project is whether more flexible tree based classifiers combined with systemic hyperparameter tuning, can improve NF1 genotype prediction and generalized compared to the logistic regression model used by Tomkinson et al.

1.1 Project Goals and Contributions

My goal was to reproduce the original logistic regression pipeline from cell painting to classification of the cells collected. NF1 wild type and NF1 knockout (cells with the NF genetic disease) were plated on 96 well plates and stained with a four color cell painting panel to label the different structures in the cell. The cells were then imaged with a microscope and pushed through the cell painting pipeline to create tabular representations of the cell's morphological features. The original paper leveraged a logistic regression classifier to predict the cell type of the image. My goal was to extend the machine learning classifier to leverage Random Forest and XGBoost while leaving the cell painting pipeline which doesn't have AI significance alone.

Additionally, the original paper does not include any standard frameworks for hyperparameter tuning. I wanted to see if a systematic approach leveraging Optuna [2] would provide better results.

This project is high risk because stronger nonlinear models may overfit the limited Schwann cell dataset and still fail to generalize across plate or lab effects, so there is no guarantee that RF or XGBoost will outperform logistic regression. Additionally, there's no guarantee that the original github repo will run in my local environment. My theory is that more flexible models and more systemic hyperparameter tuning will lead to better results.

2. Related Work

2.1 Cell Painting and Morphology based Machine Learning

In the original study by Tomkinson et al[1], the authors developed an image based pipeline followed by a logistic regression model to distinguish the NF1 mutant from the wild type Schwann cells. In this project, I first reproduced their end to end pipeline leveraging their code in their github repository and their data. I then extended their analysis by replacing their logistic regression model with a random forest classifier to test whether a non linear model can better capture the relationship between the morphology features and the NF1

genotype. In addition to random forests, I also evaluated XGBoost and introduced an Optuna based hyperparameter optimization across all models, while keeping the Cell Painting and feature extraction pipeline fixed.

If this is the case, this would allow for more effective screening of therapeutic agents that could treat neurofibromatosis.

2.2 Biological Machine Learning

2.2.1 Logistic Regression. Logistic regression is a classic baseline for machine learning where there is a categorical decision such as the absence or presence of a disease. This model is frequently used to determine an outcome based on a set of patient features. For example, a logistic regression can take a set of patient features and predict the likelihood of a disease such as kidney failure. [3]

Logistic regression models the odds of a binary outcome as a linear function of the input features passed through a sigmoid to obtain a probability. The C value is a critical value in regularization. The goal is to provide a C value that provides good test accuracy avoiding overfitting and underfitting. C is the inverse of the regularization strength and avoids fitting noise in the training data by penalizing large coefficients. [4]

2.2.3 Random Forests. Random Forest (RF) is an ensemble learning algorithm that constructs a forest of decision trees using bootstrap aggregating and random feature selection to improve predictive accuracy while reducing overfitting. In biomedical signal processing and clinical data analysis, RF models have demonstrated robust performance for tasks such as disease classification and biochemical marker prediction. [5] The balance of high flexibility and reduced overfitting has made RF a popular choice in healthcare AI.

There are key hyperparameters that one can update to establish a balance between model flexibility and overfitting risk. These key hyperparameters include `n_estimators` (number of trees), `max_depth` (maximum tree depth), `min_samples_leaf` (minimum samples per leaf) and `max_features` (features considered per split). Increasing `n_estimators` generally improves the stability of the ensemble while constraining each trees depth and requiring a larger leaf size will prevent overfitting. [6]

2.2.2 XGBoost. XGBoost is a tree based ensemble learning method that's leveraged in many studies of biomedical data due to its predictive performance and its flexibility. XGBoost has been successfully applied to predict treatment responses and adverse events from tabular datasets based on clinical trials. XGBoost models have the ability to capture nonlinear interactions that linear models might miss.[7] This makes XGBoost an ideal candidate for detecting NF1 from the Cell Painting features.

XGBoost is a gradient boosted decision tree algorithm that builds an ensemble of shallow trees that each corrects the errors of the previous tree. Unlike random forests, which fit trees independently and average them, XGBoost adds trees sequentially using gradient information from the loss function, which can capture non linear interactions but also increases the risk of overfitting if not regularized [8]

There were a few hyperparameters that dominated my better performing models. Those hyper parameters were the number of trees (`n_estimators`) and their maximal depth (`max_depth`). The

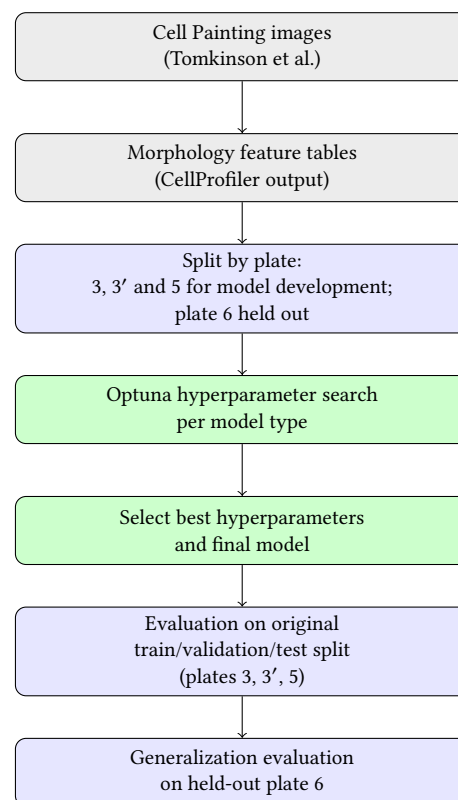


Figure 1: Overview of the NF1 Schwann cell modeling workflow. Gray boxes indicate upstream steps from Tomkinson et al. reused without modification, green boxes highlight the new hyperparameter optimization components introduced in this study, and blue boxes denote model specific training and evaluation run for each classifier.

increased number of trees and larger number of boosting rounds allow the model to fit more complex patterns without the risk of overfitting with the appropriate regularization.[9]

3.0 Methodology

I followed the original NF1 Schwann cell Cell Painting pipeline described by Tomkinson et al [1] as shown in Figure 1. For model development, I pooled single cells from plates 3, 3' and 5 and used the existing train, validation and test indices provided in the original repositories. Additionally, I added an Optuna based hyperparameter search to provide a more efficient way of finding the best hyperparameters for my model. Plate 6 was held out entirely and saved later for generalizability analysis. All models were trained to classify individual cells as NF1-null (positive class) versus NF1 wild type (negative class)

3.1 Code Cleanup

I began by cloning the `nf1_cellpainting_data` repository to run CellProfiler and Cytotable for feature extraction and importing cell data into databases. I updated these notebooks only where necessary to fix paths, data loading logic, and environment variables.

I relied primarily on the Git LFS hosted feature tables released by the authors and used those for downstream analysis rather than rerunning the feature extraction, as I felt most improvements could be made to the machine learning portion of this study.

I then forked the `nf1_schwann_cell_morphology_signature` repository and updated the notebook paths to run the repository in my local environment. The original paper did not leverage any framework for tracking and analyzing hyperparameters, and I felt I could improve upon the existing model by making the hyperparameters explicit and tunable. I also cleaned up the code to replace hard-coded paths with configurable local paths.

3.2 Optuna

Hyperparameter optimization was performed with Optuna.[2] I took advantage of the library's ability to persist training runs to SQLite. I created a dedicated Optuna study for each approach (i.e. random forest, logistic regression and XGBoost). I only add new training runs if the training runs did not already exist in the SQLite database. For each trial I performed K fold cross validation on the training set and logged training metrics along with the sampled hyperparameters. These tables allowed us to visualize hyperparameter importance to determine adjustments required in training.

3.3 Approach

My modeling workflow followed four steps:

- Train each classifier with Optuna-based hyperparameter tuning on Plates 3, 3' and 5.
- Evaluate the models on the held out test cells from these plates.
- Assess generalizability by applying the final models to an unseen Plate 6.
- Aggregate and visualize confusion matrices.

3.4 Train Models

3.4.1 Logistic Regression. The logistic regression was a reproduction of the original paper's model. I leveraged scikit learn's `LogisticRegression`. NF1 null cells (C04) were encoded as a positive class. Within each Optuna trial, I sampled a single regularization strength C from a continuous range. I performed 5 fold cross validation. For each fold I computed accuracy. After all of the trials completed, I selected the C value corresponding to the best accuracy.

3.4.2 Random Forest. I used scikit learn's `RandomForestClassifier` and I varied the number of trees, maximum depth, minimum samples per leaf and the number of features considered at each split while keeping the other parameters fixed. With each Optuna trial I sampled one combination from the grid and trained the model using 5 fold cross validation. I selected the hyperparameters that maximized the global cross validated AUC per Bradley's approach [10].

3.4.3 XGBoost. I used XGBoost's `XGBClassifier` and defined a hyperparameter grid over the number of trees, maximum tree depth, learning rate, subsampling fractions for rows and features, minimum child light, and L1/L2 regularization. For each Optuna trial, I sampled one combination from this grid and trained the model

using 5 fold cross-validation. I selected the hyperparameters that maximized the global cross-validated AUC, following Bradley's approach [10].

3.5 Evaluate Model

For each trained model, I evaluated performance on the test split from plates 3, 3' and 5 using the same indices as the original study. I computed accuracy, F1 score, precision and recall with the NF1 null cells. Additionally, I reproduced the confusion matrices in the original paper for all trained models to evaluate performance on the Test and Train data.

3.6 Assess Generalizability

I evaluated the trained models on the held out Plate 6 which was not used during training. Plate 6 contained Schwan cells from two institutions. I summarized Plate 6 performance by computing institution specific accuracies on both shuffled and non-shuffled features to ensure I was learning morphology and not artifacts.

4.0 Results

Across all three models, the combination of the feature set from the Cell Painting feature set and an Optuna based hyperparameter search produced classifiers that reliably distinguish between NF1-null (C04) from NF1 wild type (A3) Schwann cells on the held out test plates. Logistic regression kept a strong baseline (test accuracy .79), XGBoost had a slightly higher test accuracy (.80) and random forest performed slightly worse.

Hyperparameter analysis confirms that the Optuna studies were run to saturation for all three models (725 trials for logistic regression, 100 for random forest and 100 for XGBoost). It was determined that a small number of hyperparameters had the most impact on performance. Those hyperparameters were `n_estimators/max_depth` for XGBoost and `max_features` for Random Forest. I ran the same shuffled baseline analysis as the original paper (.5 accuracy) to confirm that the classifiers are learning morphology rather than artifacts.

4.1 Hyperparameter Analysis

I used Optuna studies to systematically explore hyper parameters for each model type.

Random Forest. For the random forest, I tuned `n_estimators`, `max_depth`, `min_samples_leaf` and `max_features` using mean ROC AUS as the Optuna objective. Hyper parameter analysis shows that `max_features` overwhelmingly dominates model performance (importance .96) with `n_estimators` contributing modestly as shown in Figure 2.

For XGBoost, I sampled a richer set of hyper parameters: `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `min_child_light`, `gamma`, `lambda` and `alpha`. In the Optuna importance ranking, `n_estimators` and `max_depth` are the two dominant factors in importance as shown in Figure 3.

4.2 Evaluation

After selecting the best hyperparameters from the Optuna Studies, I retrained each model on the held out test set using the authors

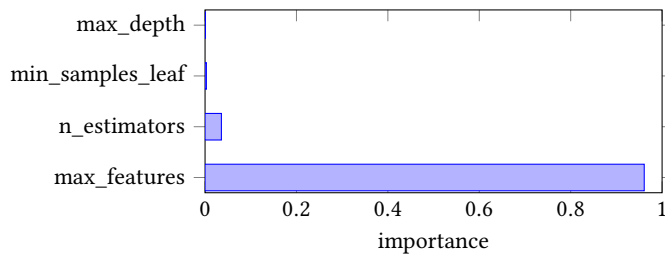


Figure 2: Max_features is the most important hyperparameter for random forest.

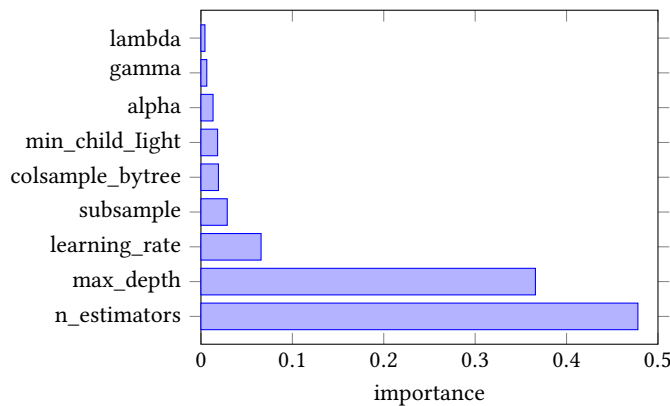


Figure 3: Max_depth and n_estimators are the most important features.

original test splits. XGBoost showed a slightly higher average metrics across all plates for the test split. Table 1, 2 and 3 show the results across the plates I leveraged for the Test split on the data leveraged for training.

I was able to reproduce the confusion matrix from the study with my version of logistic regression as shown in Figures 7 - 7. I searched slightly longer for hyperparameters than the original study which result in a tiny overfitting of the training set and slightly worse performance on the test set.

Random Forest and XGBoost had a tendency to perform better with the training fit to the extent of overtraining. XGBoost had a small improvement on the test set accuracy (.80% accuracy vs .79% accuracy for logistic regression) as shown in Figures 8 - 11.

Table 1: Logistic Regression metrics for the test evaluation across plates.

Plate	f1_score	precision	recall	accuracy
all_plates	0.790	0.786	0.794	0.789
Plate_3	0.847	0.840	0.855	0.846
Plate_3_prime	0.817	0.801	0.834	0.813
Plate_5	0.630	0.642	0.619	0.637

Table 2: XGBoost metrics for the test evaluation across plates.

Plate	f1_score	precision	recall	accuracy
all_plates	0.801	0.802	0.799	0.801
Plate_3	0.880	0.888	0.873	0.881
Plate_3_prime	0.827	0.763	0.903	0.811
Plate_5	0.578	0.630	0.534	0.610

Table 3: Random Forest metrics for the test evaluation across plates.

Plate	f1_score	precision	recall	accuracy
all_plates	0.764	0.749	0.779	0.759
Plate_3	0.836	0.823	0.851	0.834
Plate_3_prime	0.783	0.741	0.829	0.770
Plate_5	0.576	0.583	0.570	0.581

4.3 Generalization

I evaluated all three classifiers on Plate 6 which contain Schwann cells profiled at two institutions (MGH and iNFixion) that were not used in model training or hyperparameter tuning. For each model, I computed per institution accuracy as shown in Tables 4, 5 and 6.

MGH performed poorly across all models. Note in tables 4, 5 and 6 that the MGH performance performed no better than the shuffled control.

iNFixion saw the best performance with the classic logistic regression with an accuracy of .688 while keeping the shuffled baseline analysis at .5. iNFixion had an accuracy of .630 but it's likely simply learning artifacts as the shuffled features obtained an accuracy of .625. XGBoost had and a similar accuracy while showing minimal evidence of learning artifacts with shuffled features showing an accuracy of .471

Table 4: Plate 6 accuracies for logistic regression.

Metadata_Institution	data_type	accuracy
MGH	final	0.495
MGH	shuffled	0.500
iNFixion	final	0.688
iNFixion	shuffled	0.502

Table 5: Plate 6 accuracies for XGBoost.

Metadata_Institution	data_type	accuracy
MGH	final	0.524
MGH	shuffled	0.466
iNFixion	final	0.629
iNFixion	shuffled	0.471

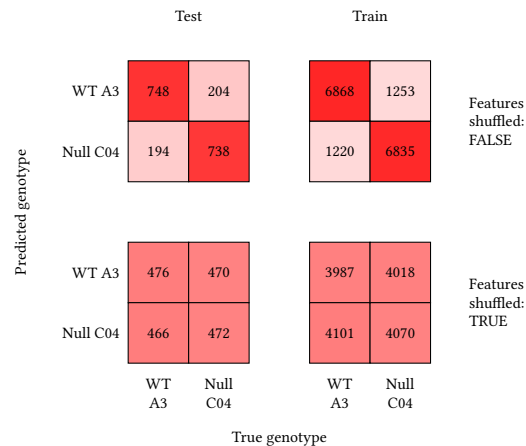


Figure 4: Logistic regression confusion counts for NF1 genotype classification on train and test cells, with and without feature shuffling.

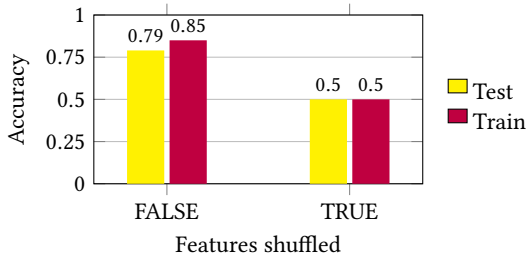


Figure 5: Represents my replication of the study for logistic regression

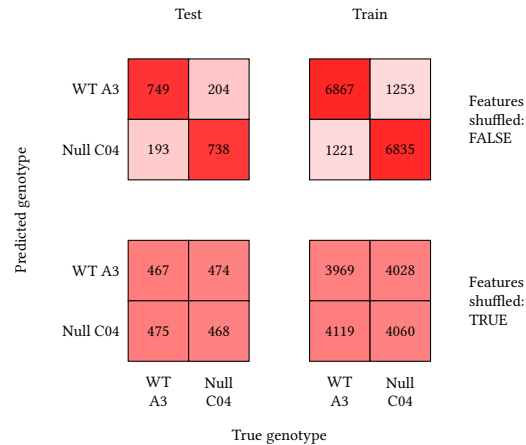


Figure 6: Logistic regression confusion counts from the original study for NF1 genotype classification on train and test cells, with and without feature shuffling.

Table 6: Plate 6 accuracies for Random Forest.

Metadata_Institution	data_type	accuracy
MGH	final	0.592
MGH	shuffled	0.577
iNFixion	final	0.630
iNFixion	shuffled	0.625

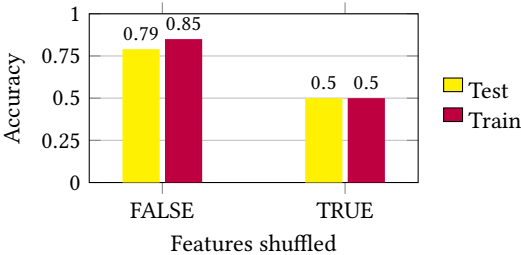


Figure 7: Logistic regression results from the original study.

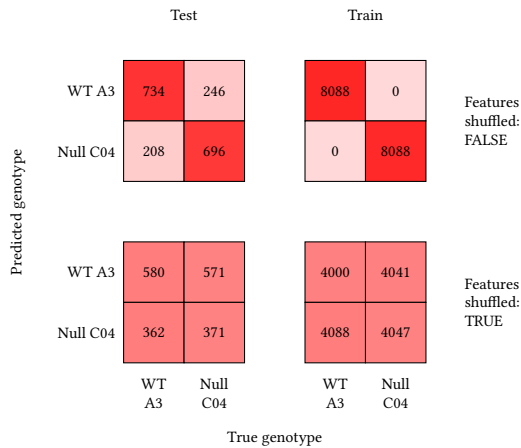


Figure 8: Random forest confusion counts from the original study for NF1 genotype classification on train and test cells, with and without feature shuffling.

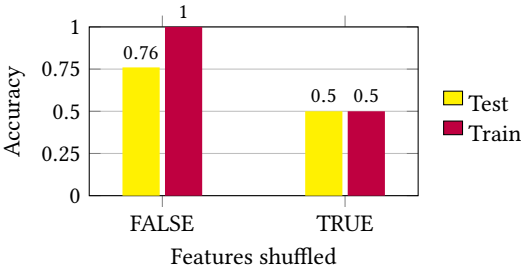


Figure 9: Represents the Random Forest Version of the study

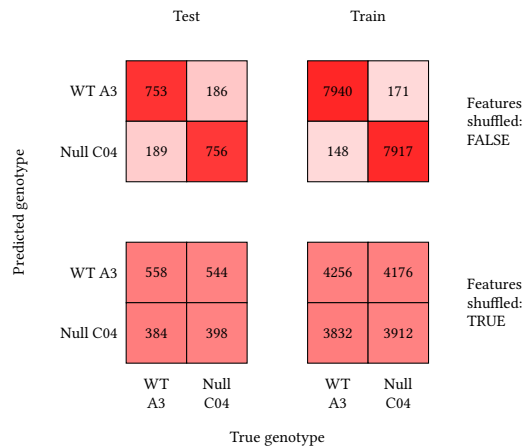


Figure 10: XGBoost confusion counts from the original study for NF1 genotype classification on train and test cells, with and without feature shuffling.



Figure 11: Represents the XGBoost Version of the study

Conclusion

In this study, I reproduced the NF1 Schwann Cell painting pipeline from Tomkinson et al [1] and extended it with the Optuna hyperparameter tuning framework. Additionally, I added the Random Forest and XGBoost model families. Logistic regression reproduced the original study's test accuracy of .79

Generalizability experiments on plate 6 (which included the Schwann cells from the two institutions not seen during training) for logistic regression and XGBoost showed above chance accuracies at iNFixion while performing much worse for MGH.

The findings show that the simple logistic regression offers strong performance for NF1 morphology classification. Random Forest overfitted the training data and did not offer any gains for the test or validation data. XGBoost showed incremental gains but introduced the risk of overfitting on the training data.

Logistic regression performs better as it's not overtraining on techniques learned from the labs that produced plates 3, 3' and 5. Logistic regression still had challenges generalizing to the MGH data set.

Future work should entail a better understanding of Plate 6 and the features that contribute to the model. Try models that explicitly correct for plate or institution effects and evaluate additional NF1 Schwann lines. Future work should also include further scrutinizing

studies successful in biochemical marker prediction to determine if there is an opportunity to port their techniques to NF1 Schwann cells.

References

- [1] J. Tomkinson et al., "High-content microscopy and machine learning characterize a cell morphology signature of NF1 genotype in Schwann cells," Sept. 16, 2024, Systems Biology. doi: 10.1101/2024.09.11.612546.
- [2] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A Next-generation Hyperparameter Optimization Framework," in Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, Anchorage AK USA: ACM, July 2019, pp. 2623–2631. doi: 10.1145/3292500.3330701.
- [3] Y. Hua, T. S. Stead, A. George, and L. Ganti, "Clinical Risk Prediction with Logistic Regression: Best Practices, Validation Techniques, and Applications in Medical Research," Acad. Med. Surg., Mar. 2025, doi: 10.62186/001c.131964.
- [4] M. Sikander, "Logistic Regression: A Simple Guide to Intuition and Implementation in Python," Towards AI.
- [5] I. V. Pantic, J. Paunovic Pantic, S. Valjarevic, P. R. Corridon, and N. Topalovic, "Artificial intelligence – based approaches based on random forest algorithm for signal analysis: Potential applications in detection of chemico - biological interactions," Chem. Biol. Interact., vol. 418, p. 111624, Sept. 2025, doi: 10.1016/j.cbi.2025.111624.
- [6] H. Singh, "Introduction to Random Forest and its Hyperparameters," Analytics Vidhya. [Online]. Available: <https://analyticsvidhya.com/blog/2021/03/introduction-to-random-forest-and-its-hyper-parameters/>
- [7] M. Wiens, A. Verone-Boyle, N. Henscheid, J. T. Podichetty, and J. Burton, "A Tutorial and Use Case Example of the XGBoost Artificial Intelligence Algorithm for Drug Development Applications," Clin. Transl. Sci., vol. 18, no. 3, p. e70172, Mar. 2025, doi: 10.1111/cts.70172.
- [8] J. Crossman-Smith, "XGBoost Explained: A Beginner's Guide," Low Code for Data Science. [Online]. Available: <https://medium.com/low-code-for-advanced-data-science/xgboost-explained-a-beginners-guide-095464ad418f>
- [9] A. Jain, "XGBoost Parameters Tuning," Analytics Vidhya. [Online]. Available: <https://analyticsvidhya.com/blog/2016/03/complete-guide-parameter-tuning-xgboost-with-codes-python/>
- [10] A. P. Bradley, "The use of the area under the ROC curve in the evaluation of machine learning algorithms," Pattern Recognit., vol. 30, no. 7, pp. 1145–1159, July 1997, doi: 10.1016/S0031-3203(96)00142-2.
- [11] Y. Lu, M. Xu, X. Chen, H. Xu, N. Sun, K. E. Iisgerber, and R.-Y. Bai, "Neurofibromatosis Type 1: Genetic Mechanisms and Advances in Therapeutic Innovation," Cancers, vol. 17, no. 23, p. 3788, 2025, doi: 10.3390/cancers17233788. Available: <https://www.mdpi.com/2072-6694/17/23/3788>
- [12] P. M. K. de Blank, A. M. Gross, S. Akshintala, J. O. Blakeley, G. Bollag, A. Cannon, E. Dombi, J. Fangusaro, B. D. Gelb, D. Hargrave, A. R. Kim, L. J. Klesse, M. Loh, S. Martin, C. Moertel, R. Packer, J. M. Payne, K. A. Rauen, J. J. Rios, et al., "MEK inhibitors for neurofibromatosis type 1 manifestations: Clinical evidence and consensus," Neuro-Oncology, vol. 24, no. 11, pp. 1845–1856, 2022, doi: 10.1093/neuonc/noac165. Available: <https://academic.oup.com/neuro-oncology/article/24/11/1845/6631282>